

Software Requirements Specification (SRS)

Project: UniConnect (CSE Departmental Hub)

Lab 2: Problem Definition + SRS v1

1. Document Control

1.1 Metadata

Field	Details
Team	Team RAAS
Project Name	UniConnect (CSE Departmental Hub)
Repository URL	https://github.com/ruhul1845/UniConnect
Version	1
Status	Draft
Date	March 08, 2026

1.2 Authors and Reviewers

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1.3 Revision History

Version	Date	Author	Changes
0.1	07-Mar-2026	Development Team	Initial draft — SRS v1
1	08-Mar-2026	Development Team	Added CR upload types: slides, books, mid-term & final question papers; new FRs, user stories, use case, and test cases

2. Problem Definition

2.1 Problem Statement

UniConnect addresses the fragmented digital infrastructure of the Computer Science & Engineering (CSE) department. Currently, academic resources are scattered across informal Google Drive links, Facebook groups, and physical notice boards — creating inefficiency, missed communications, and safety gaps.

The platform consolidates academic resources, peer-to-peer commerce, off-campus housing, and emergency safety protocols into a single verified, secure, and searchable application — acting as the 'Departmental Brain' for the CSE department.

2.2 Target Users

User Group	Description	Main Needs
CSE Students	Undergraduate/postgraduate students of the CSE department	Access resources, buy/sell items, find housing, trigger SOS
Class Representatives (CR)	Student representatives per batch	Upload lab tasks, slides, books, question papers, project showcases, course materials
Administrators	Department faculty/staff	Moderate content, manage user verification, view analytics
External Buyers/Renters	Individuals outside CSE browsing listings	Limited access — marketplace browsing only

2.3 Goals and Non-Goals

Goals:

- Centralize academic resources (slides, question papers, books, lab sheets) in a searchable repository.
- Allow CRs to upload all resource types: lecture slides, reference books (PDF), previous year mid-term and final exam papers.
- Provide a secure peer-to-peer marketplace for CSE-specific items (laptops, Arduino kits, textbooks).
- Enable off-campus housing discovery with roommate matching based on class schedules.
- Implement a real-time safety system (Panic Button + GPS SOS broadcast) for student safety.
- Enforce identity verification via university email and Facebook account linking.

Non-Goals:

- UniConnect will NOT process financial transactions or handle payments between users.
- UniConnect will NOT extend beyond the CSE department to other faculties in v1.
- UniConnect will NOT replace the official university LMS (Learning Management System).
- UniConnect will NOT provide academic grading or assignment submission features

3. Scope and Context

3.1 In Scope

- Module A: Academic Resource & Lab Archive — batch-wise repository, question bank (mid-term & final), lecture slides, reference books (PDF), lab central, project showcase, bilingual search.
- Module B: CSE Marketplace (Dorm-to-Dorm) — item listings with categories (hardware, books, software licenses), seller contact gating.
- Module C: To-Let & Accommodation Finder — off-campus housing listings, roommate matching, availability toggles.
- Module D: Safety & Verified Social — SOS panic button with GPS broadcast, proctorial directory, Facebook Auth identity verification.
- REST API backend with JWT authentication and WebSocket support for real-time alerts.
- Mobile-responsive React/Next.js frontend.

3.2 Out of Scope

- In-app payment processing or escrow services.
- Integration with other university departments or faculties.
- Native mobile apps (iOS/Android) — web-responsive only in v1.
- Automated course scheduling or timetable generation.
- AI-based recommendation engines for resources or housing.

3.3 Assumptions and Constraints

Technical Constraints:

- Backend: Node.js / Express REST API with PostgreSQL database.
- File storage: AWS S3 or Cloudinary for PDFs and images.
- Real-time: WebSocket server (Socket.io) for SOS broadcast.
- Authentication: JWT tokens; Facebook OAuth for identity verification.
- Uploaded files must be PDF format; maximum file size 25 MB per upload.

Organizational Constraints:

- Only verified university-domain email addresses can register.
- Marketplace and housing contact info visible to verified CSE members only.
- Content moderation is required before any resource (slides, books, question papers, lab sheets) goes live.
- Only users with the CR role may upload academic resources.

Timeline Constraints:

- Lab 2 SRS submission: March 2026.
- MVP delivery expected within one academic semester.

4. Stakeholders and Roles

4.1 Stakeholders

Stakeholder	Interest	Priority
CSE Students	Access to resources, safe marketplace, housing, emergency tools	High
Class Representatives	Upload and manage all departmental content types	High
Department Administration	Content governance, student safety oversight	High
University IT Dept.	Infrastructure, data security compliance	Medium
Parents / Guardians	Student safety via SOS alerts	Medium

4.2 User Roles and Permissions

Role	Description	Permissions
Student (Verified)	Registered CSE student with verified university email	Browse/download resources (slides, books, question papers, lab sheets), post/view marketplace, post/view housing, trigger SOS
Class Representative (CR)	Designated batch representative	All student permissions + upload lab sheets, lecture slides, reference books (PDF), mid-term question papers, final exam question papers, project showcases
Admin	Department faculty or appointed moderator	All CR permissions + moderate/delete content, manage user verification, view reports
Guest (Unverified)	Non-authenticated visitor	View public landing page only; no resource/contact access

5. Functional Requirements

Requirement ID format: FR-UC-###

Req ID	Feature	Requirement Statement	Priority	Acceptance Criteria
FR-UC-001	User Registration	System shall allow registration using university-domain email only	M	Non-university email rejected with error message
FR-UC-002	FB-Auth Integration	System shall require mandatory Facebook account linking for identity verification	M	Login fails without linked FB profile
FR-UC-003	Batch Repository	System shall organize resources by Batch > Semester > Course hierarchy	M	Navigation drill-down returns correct files
FR-UC-004	Question Bank — Mid-Term	System shall provide a searchable database of previous year mid-term exam papers with tag filtering by course code and year	M	Searching a course code returns only mid-term papers for that course; year filter works correctly
FR-UC-004B	Question Bank — Final	System shall provide a searchable database of previous year final exam papers with tag filtering by course code and year	M	Searching a course code returns only final papers; year and exam-type filters work correctly
FR-UC-005	Lab Central	System shall store and serve lab task sheets per course; only CRs may upload	M	CR uploads PDF; students can download after admin approval
FR-UC-005B	Lecture Slides Upload	System shall allow CRs to upload lecture slide decks (PDF) tagged by Batch, Semester, Course, and Lecture number	M	Uploaded slide PDF appears under correct course within 1 minute after admin approval; lecture number tag is searchable
FR-UC-005C	Reference Books Upload	System shall allow CRs to upload reference book	M	Uploaded book PDF appears under correct course;

		PDFs tagged by course code, author, and edition		author/edition metadata is displayed
FR-UC-005D	Mid-Term Question Paper Upload	System shall allow CRs to upload previous year mid-term question papers tagged with course code, year, and exam type = Mid-Term	M	Uploaded paper appears in the Question Bank under Mid-Term filter for the correct course and year
FR-UC-005E	Final Exam Question Paper Upload	System shall allow CRs to upload previous year final exam question papers tagged with course code, year, and exam type = Final	M	Uploaded paper appears in the Question Bank under Final filter for the correct course and year
FR-UC-006	Project Showcase	System shall display a 'Wall of Fame' linking student projects to GitHub repositories	S	Valid GitHub URL required on submission
FR-UC-007	Bilingual Search	System shall support searching course materials in both English and native script	S	Native-language query returns same results as English
FR-UC-008	Marketplace Listing	System shall allow verified users to post items with name, price, condition, and images	M	Non-logged-in users cannot view seller contact info
FR-UC-009	Marketplace Categories	System shall filter marketplace items by Hardware, Software Licenses, and Academic Books	M	Category filter returns only items of that type
FR-UC-010	Housing Listings	System shall allow users to post off-campus housing listings with rent, type, and location	M	Listing appears on map with correct coordinates
FR-UC-011	Roommate Matching	System shall filter roommate requests by synchronized CSE lab/class schedules	S	CSE-Only filter hides non-CSE listings

FR-UC-012	Availability Toggle	System shall provide real-time Available/Occupied status for housing listings	M	Toggle update reflected within 5 seconds
FR-UC-013	SOS Panic Button	System shall broadcast user GPS coordinates to nearest active CSE students on 3-second hold	M	Alert reaches active users within 2 seconds of trigger
FR-UC-014	Proctorial Directory	System shall provide a one-tap call list for Campus Security, Medical, and Dept. Head	M	Directory loads within 2 seconds; calls initiate correctly
FR-UC-015	Content Moderation	System shall allow Admins to review, approve, or delete all uploaded content including slides, books, and question papers	M	Unapproved content not visible to students
FR-UC-016	Upload Type Tagging	System shall enforce mandatory metadata tagging on every upload: resource type (Slide/Book/Mid-Term/Final/Lab), course code, batch, semester	M	Upload rejected with validation error if any required tag is missing
FR-UC-017	Duplicate Upload Detection	System shall warn CR when an identical file name + course code + year combination already exists in the repository	S	Duplicate warning shown; CR can choose to overwrite or cancel

6. User Stories and Use Cases

6.1 User Stories

Story ID	As a...	I want...	So that...	Acceptance Criteria
US-UC-001	CSE Student	to browse resources by batch and course	I can find relevant slides quickly	Resource tree navigable in ≤ 3 clicks to file
US-UC-002	CSE Student	to search for past mid-term exam papers by course tag and year	I can prepare for mid-terms efficiently	Tag + year filter returns only matching mid-term papers
US-UC-002B	CSE Student	to search for past final exam papers by course tag and year	I can prepare for finals efficiently	Tag + year filter returns only matching final exam papers
US-UC-003	CSE Student	to download lab sheets for my enrolled courses	I can complete labs without asking seniors	Download link active; file opens correctly
US-UC-003B	CSE Student	to download lecture slides for my courses	I can study without relying on informal sharing groups	Slide PDF downloads correctly; tagged with correct lecture number
US-UC-003C	CSE Student	to download reference book PDFs for my courses	I have access to course books without buying them	Book PDF downloads with author and edition shown
US-UC-004	CSE Student	to view student projects with GitHub links	I can learn from senior projects	Each project card shows title, description, and working GitHub link
US-UC-005	CSE Student	to list a used laptop for sale	I can sell to trusted department peers	Listing visible to verified users only; seller contact hidden from guests
US-UC-006	CSE Student	to filter marketplace by hardware category	I find only relevant items	Category filter shows only hardware items
US-UC-007	CSE Student	to post a housing listing with my flat's location	Batchmates can find accommodation near campus	Map pin renders at correct coordinates
US-UC-008	CSE Student	to find a roommate with synchronized class schedule	I avoid schedule conflicts with my roommate	CSE-Only filter hides non-department users
US-UC-009	CSE Student	to trigger an SOS alert in an emergency	Nearby students and security are notified immediately	Alert dispatched within 2 seconds; GPS coordinates accurate

US-UC-010	CSE Student	to access emergency contact numbers with one tap	I can reach security or medical staff fast	Directory loads within 2 seconds; tap initiates call
US-UC-011	Class Rep (CR)	to upload lab task sheets for my batch	Students have structured access to lab materials	Uploaded file appears under correct batch/course within 1 minute
US-UC-011B	Class Rep (CR)	to upload lecture slides tagged by course and lecture number	Students can access slides in an organised, searchable way	Slide appears under correct course; lecture number tag is searchable
US-UC-011C	Class Rep (CR)	to upload previous year mid-term question papers tagged with course code and year	Students can find past mid-term papers quickly	Paper appears in Question Bank under Mid-Term filter for correct course and year
US-UC-011D	Class Rep (CR)	to upload previous year final exam question papers tagged with course code and year	Students can find past final papers quickly	Paper appears in Question Bank under Final filter for correct course and year
US-UC-011E	Class Rep (CR)	to upload reference book PDFs with author and edition metadata	Students have one-stop access to course books	Book appears under correct course; metadata (author, edition) is displayed correctly
US-UC-012	Class Rep (CR)	to submit our team's project to the showcase	Our work is visible on the departmental Wall of Fame	Project rejected if GitHub URL format is invalid
US-UC-013	Admin	to review and approve uploaded resources before publication	Only verified, quality content reaches students	Pending items listed in moderation queue; approve/reject works
US-UC-014	Admin	to revoke access for a user who violates community rules	The platform remains safe and trustworthy	Revoked user cannot log in; their posts hidden
US-UC-015	Unverified User	to be blocked from seller contact info	Seller privacy is protected from outsiders	Contact info replaced with login prompt for unverified users
US-UC-016	CSE Student	to search for course materials in Bengali script	I can find resources regardless of which language I search in	Native-language query returns same results as English equivalent

6.2 Critical Use Cases

UC-01: Student Accesses Academic Resource

Field	Detail
Preconditions	User is logged in with verified university email
Main Flow	1. User navigates Home > Archive Tab. 2. Selects Batch > Semester > Course. 3. Selects resource type tab: Slides Books Question Papers (Mid-Term / Final) Lab Sheets. 4. Browses available PDFs. 5. Clicks download to save file.
Alternate Flow	If no resources exist for the course, system shows 'No resources uploaded yet' placeholder.
Exception Flow	If file URL is broken, system shows error and logs the issue to admin queue.
Postconditions	File is downloaded; download count incremented in DB.

UC-02: Student Triggers SOS Panic Alert

Field	Detail
Preconditions	User is logged in, GPS permissions granted, active internet connection
Main Flow	1. User holds Panic Button for 3 seconds from any screen. 2. Emergency overlay activates. 3. System broadcasts GPS coordinates via WebSocket to nearest active CSE users. 4. SOS notification delivered within 2 seconds.
Alternate Flow	If GPS unavailable, system broadcasts last known location with a warning flag.
Exception Flow	If WebSocket connection fails, system attempts HTTP fallback push notification.
Postconditions	Alert logged with timestamp and coordinates; nearby users notified.

UC-03: CR Uploads Academic Resource [Updated — covers all upload types]

Field	Detail
Preconditions	User is logged in with CR role; file is PDF; file size $\leq$ 25 MB
Main Flow	1. CR navigates to Upload section. 2. Selects resource type from dropdown: • Lab Sheet — tags: Batch, Semester, Course • Lecture Slide — tags: Batch, Semester, Course, Lecture No. • Reference Book — tags: Course Code, Author, Edition • Mid-Term Question Paper — tags: Course Code, Year, Exam Type = Mid-Term • Final Exam Question Paper — tags: Course Code, Year, Exam Type = Final

	3. System validates all required tag fields are filled. 4. System checks for duplicate (same file name + course code + year). 5. CR uploads PDF to cloud storage. 6. Admin receives moderation notification with resource type and tags.
Alternate Flow	If file exceeds 25 MB size limit, system rejects upload and shows allowed limit. If duplicate detected, CR is warned and can choose to overwrite or cancel.
Exception Flow	If cloud storage fails, system queues upload and retries within 5 minutes; CR sees a 'Upload queued' notification.
Postconditions	File stored in cloud under correct resource-type category; pending admin approval; not visible to students until approved.

UC-04: Student Posts Marketplace Item

Field	Detail
Preconditions	User is logged in with verified university email; user has an item to list
Main Flow	1. User navigates Home > Market Tab > 'Post Item'. 2. Fills in item name, price, condition, category, and uploads images. 3. System validates price > 0 and category is valid ENUM. 4. Listing saved with status = 'Available'; visible to all verified users. 5. Seller's FB Messenger link gated — only shown to verified users.
Alternate Flow	If image upload fails, user can submit listing without images and add later.
Exception Flow	If price field is 0 or negative, system shows validation error: 'Price must be greater than 0'.
Postconditions	Item appears in marketplace; unverified users see login prompt instead of seller contact info.

UC-05: Admin Moderates Uploaded Content

Field	Detail
Preconditions	Admin is logged in; at least one resource or project is pending in moderation queue
Main Flow	1. Admin navigates to Admin Dashboard > Moderation Queue. 2. Reviews pending item (file preview, uploader ID, resource type, course tag, year). 3. Clicks 'Approve' to publish or 'Reject' with reason. 4. System moves approved item to student-facing archive under the correct resource-type tab; rejected item is flagged and uploader notified.
Alternate Flow	If queue is empty, dashboard shows 'No items pending review' message.

Exception Flow	If admin accidentally approves wrong item, Admin can use 'Remove' action post-approval to hide it.
Postconditions	Approved content visible to students under correct type tab; rejected item removed from queue; uploader email notification sent.

7. Data Requirements

7.1 Core Entities

Entity	Description	Key Fields	Constraints
USER	Registered department member	student_id (PK), email, is_verified, fb_uid	Email must be university domain; fb_uid unique
BATCH	Academic batch grouping	batch_id (PK), batch_year, department	References multiple USERS and PROJECTs
RESOURCE	Uploaded academic file (any type)	resource_id (PK), file_url, resource_type, course_code, uploader_id (FK), year, metadata (JSONB)	file_url required; resource_type ENUM; course_code indexed; uploader_id FK to USER
PROJECT	Student project showcase entry	project_id (PK), title, github_link, batch_id (FK)	github_link must match GitHub URL pattern
MARKET_ITEM	Marketplace listing item	item_id (PK), item_name, price, status, seller_id (FK)	price > 0 (Decimal); status ENUM: Available, Sold
HOUSING_POST	Off-campus housing or roommate post	post_id (PK), type, rent, location_coords, is_cse_only	type ENUM: Sublet/Flat/Roommate; is_cse_only Boolean

The **RESOURCE** entity now uses a unified model with a **resource_type** ENUM and a **JSONB** metadata field to support all upload types:

Resource Type	Required Metadata Fields	Stored Under
Lab Sheet	batch, semester, course_code	Archive > Course > Lab Sheets
Lecture Slide	batch, semester, course_code, lecture_no	Archive > Course > Slides
Reference Book	course_code, author, edition	Archive > Course > Books
Mid-Term Paper	course_code, year, exam_type=Mid-Term	Question Bank > Mid-Term
Final Exam Paper	course_code, year, exam_type=Final	Question Bank > Final

7.2 Data Dictionary

Field	Type	Required	Validation / Constraint	Example
student_id	UUID	Yes	Primary Key, auto-generated	550e8400-e29b-41d4-a716
email	VARCHAR(255)	Yes	Unique; must match @university.edu domain	john@cse.university.edu
is_verified	BOOLEAN	Yes	Default: false; set true after email verification	true

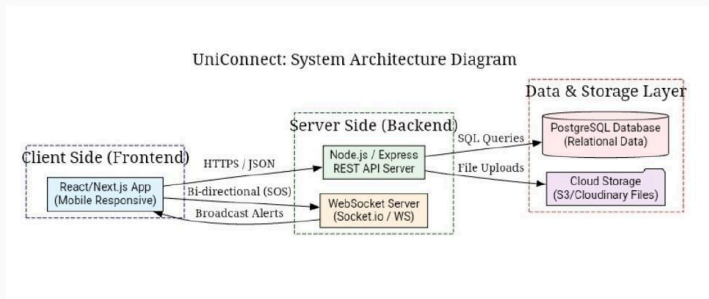
fb_uid	VARCHAR(100)	Yes	Unique; obtained from Facebook OAuth flow	fb_123456789
file_url	TEXT	Yes	Must be valid HTTPS URL (S3 or Cloudinary)	https://s3.amazonaws.com/...
resource_type	ENUM	Yes	Values: Lab_Sheet, Slide, Book, MidTerm_Paper, Final_Paper	Slide
course_code	VARCHAR(20)	Yes	Indexed; format: CSE-XXX	CSE-301
year	INTEGER	Conditional	Required for Mid-Term and Final papers; must be $\leq$ current year	2024
metadata	JSONB	Yes	Structure varies by resource_type; validated server-side	{'lecture_no': 5}
github_link	TEXT	Yes (Projects)	Must match regex: https://github.com/.*	https://github.com/user/repo
price	DECIMAL(10,2)	Yes	Must be > 0	1500.00
status	ENUM	Yes	Values: Available, Sold	Available
location_coords	JSONB	Yes (Housing)	Must contain lat (Float) and lng (Float)	{'lat': 23.81, 'lng': 90.41}
is_cse_only	BOOLEAN	Yes (Housing)	Default: false; restricts visibility to CSE users	true

7.3 Data Model

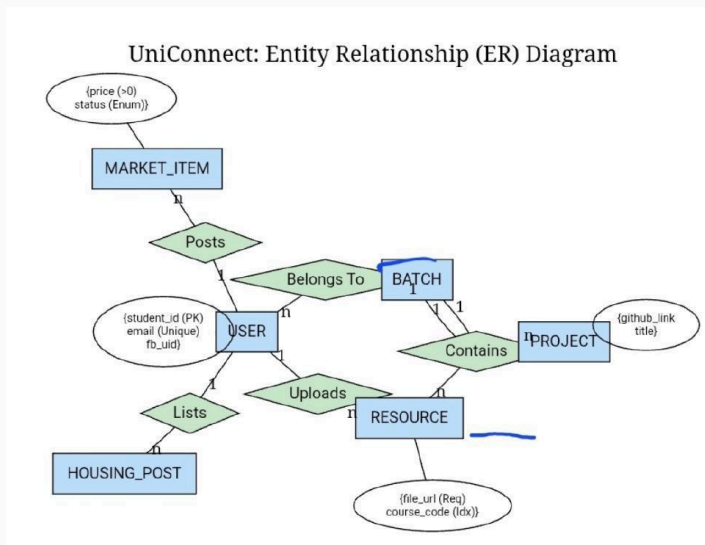
1 . System Architecture Diagram

2 . Entity Relationship (ER) Diagram

2.1 System Architecture Diagram



2.2 Entity Relationship (ER) Diagram



Key relationships:

- USER belongs to one BATCH; a BATCH contains many USERS.
- USER uploads many RESOURCES (of any type); each RESOURCE has one uploader (FK: uploader_id).
- RESOURCE has a resource_type ENUM distinguishing Lab_Sheet, Slide, Book, MidTerm_Paper, Final_Paper.
- USER posts many MARKET_ITEMS and HOUSING_POSTs.
- BATCH contains many PROJECTs; each PROJECT belongs to one BATCH.

8. UI and UX Requirements

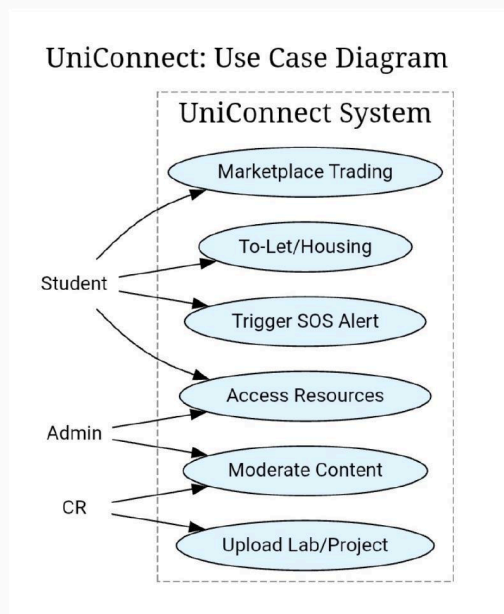
8.1 Wireframes / Navigation Flow Maps

Use Case Diagram

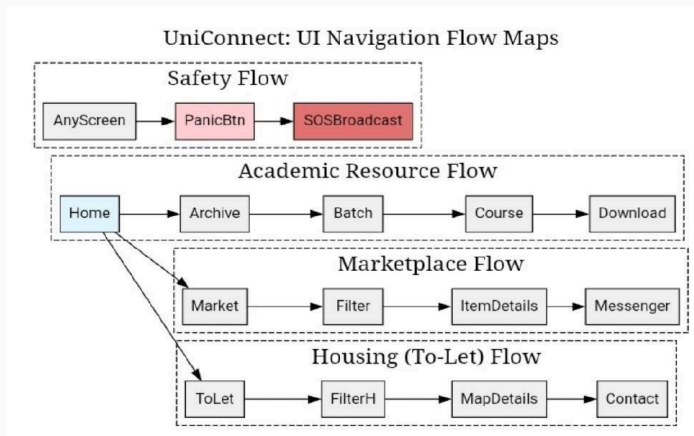
- **FB-Auth Integration:** Mandatory Facebook profile linking to ensure identity accountability.

4. Data Flow & Use Cases

4.1 Use Case Diagram



6. UI Flow Maps (Navigation)



- **Resource Flow:** Home → Archive Tab → Select Batch → Select Course → Download PDF/Open GitHub.
- **Marketplace Flow:** Home → Market Tab → Filter Category → Item Details → Contact via Messenger.
- **Housing Flow:** Home → To-Let Tab → Filter (Roommate/Flat) → View Map/

The following 5 primary flows are defined:

- Resource Flow: Home > Archive Tab > Select Batch > Select Semester > Select Course > Select Resource Type Tab (Slides | Books | Question Papers | Lab Sheets) > Download PDF.
- CR Upload Flow: Home > Upload Section > Select Resource Type > Fill Metadata Tags > Upload PDF > Await Admin Approval.
- Marketplace Flow: Home > Market Tab > Filter Category > View Item Details > Contact Seller via Messenger.
- Housing Flow: Home > To-Let Tab > Filter (Roommate / Flat) > View Map Details > Contact Lister.
- Safety Flow: Any Screen > Emergency Overlay > Hold Panic Button 3 seconds > SOS Broadcast Active.
- Admin Flow: Admin Dashboard > Moderation Queue > Review Content (type + tags) > Approve / Reject.

8.2 UX and Accessibility Baseline

- Navigation: Clear tab-based navigation with icons and labels; resource archive has sub-tabs for Slides, Books, Question Papers (Mid-Term / Final), and Lab Sheets.
- Upload UX: CR upload form dynamically shows/hides metadata fields based on selected resource type.
- Keyboard Accessibility: All core flows (search, upload, marketplace browsing) navigable via keyboard.
- Contrast and Typography: Minimum WCAG AA contrast ratio (4.5:1) for all text; minimum font size 14px.
- Responsive Design: All screens adapt for mobile (320px+), tablet (768px+), and desktop (1024px+).
- Emergency UI: SOS Panic Button persistently accessible from all screens with high-contrast red indicator.

9. Non-Functional Requirements

NFR ID format: NFR-UC-###

NFR ID	Category	Requirement	Target / Metric
NFR-UC-001	Performance	API response time for resource queries	$\leq 500\text{ms}$ for 95% of requests under normal load
NFR-UC-002	Performance	SOS alert delivery latency	Active users notified within 2 seconds of 3-second hold trigger
NFR-UC-003	Security	All API endpoints require JWT Bearer token authentication	401 Unauthorized returned for missing/invalid tokens
NFR-UC-004	Security	Seller contact info hidden from unverified users	0% exposure rate to unverified/guest users in testing
NFR-UC-005	Security	User passwords stored as hashed values	bcrypt with minimum 12 salt rounds
NFR-UC-006	Usability	New user onboarding completion rate	$\geq 80\%$ of users complete registration without external help
NFR-UC-007	Availability	Platform uptime during academic semester	$\geq 99\%$ uptime during peak hours (8 AM–10 PM)
NFR-UC-008	Scalability	System handles concurrent active users	Minimum 500 concurrent users without performance degradation
NFR-UC-009	Data Integrity	File uploads validated before storage; all types (slides, books, papers)	Rejected files logged; no corrupted files stored in S3/Cloudinary; resource_type tag validated server-side
NFR-UC-010	Compliance	User data stored in compliance with institutional data policy	No PII shared with third parties beyond FB Auth OAuth flow
NFR-UC-011	Storage	Per-user upload quota enforced to prevent cost overrun	Max 25 MB per file; max 500 MB total uploads per CR per semester

10. Risks and Open Questions

10.1 Risk Register

Risk ID	Description	Probability	Impact	Mitigation
RISK-UC-001	Facebook OAuth API changes or deprecation break FB-Auth integration	Medium	High	Abstract auth layer; maintain fallback email-only verification option
RISK-UC-002	GPS spoofing on SOS panic button triggers false emergency broadcasts	Low	High	Rate-limit SOS triggers; require confirmation on cancel; log all SOS events
RISK-UC-003	Low student adoption due to change from existing Facebook groups	High	High	Launch with CR-driven content seeding; onboarding incentives (e.g., resource access gating)
RISK-UC-004	Cloud storage cost overrun from unrestricted file uploads	Medium	Medium	Enforce file size limits (max 25 MB); quota per user per semester (NFR-UC-011)
RISK-UC-005	Non-CSE users bypass CSE-Only filter for housing/marketplace	Low	Medium	Server-side validation of is_verified and batch_id on every protected endpoint
RISK-UC-006	WebSocket server overload during mass SOS alert scenario	Low	High	Horizontal scaling with load balancer; message queue (Redis) for broadcast routing
RISK-UC-007	CRs upload incorrect resource type (e.g., final paper tagged as mid-term)	Medium	Medium	Enforce mandatory type tagging (FR-UC-016); admin reviews type + tags before approval
RISK-UC-008	Copyright concerns over uploaded book PDFs	Medium	High	CR must confirm ownership/right-to-share during upload; admin reviews for copyright issues in moderation queue

10.2 Open Questions

- Should the Panic Button require a confirmation step to prevent accidental triggers, or should speed be prioritized?
- Will the university provide an official API for student ID verification, or will manual email verification suffice?
- Should marketplace transactions support in-app messaging, or rely entirely on FB Messenger redirect?
- What is the data retention policy for expired housing listings and sold marketplace items?
- Is bilingual search (English + Bengali) required at MVP, or deferred to v2?
- Should book PDF uploads require an ISBN or library catalog reference to help detect duplicates and copyright issues?
- Should a single CR be able to upload on behalf of any batch, or only their own assigned batch?

11. Testability and Traceability

11.1 Requirement Traceability

Requirement ID	Story / Use Case	Planned Test ID
FR-UC-001	US-UC-015 — Unverified user blocked from contact info	TC-UC-001
FR-UC-002	US-UC-001 — FB-Auth mandatory registration	TC-UC-002
FR-UC-003	US-UC-001 — Batch/Semester/Course navigation	TC-UC-003
FR-UC-004	US-UC-002 — Mid-term question bank tag + year filter	TC-UC-004
FR-UC-004B	US-UC-002B — Final exam question bank tag + year filter	TC-UC-004B
FR-UC-005	US-UC-011 — CR uploads lab sheet; student downloads	TC-UC-005
FR-UC-005B	US-UC-011B — CR uploads lecture slides with lecture no.	TC-UC-005B
FR-UC-005C	US-UC-011E — CR uploads reference book with metadata	TC-UC-005C
FR-UC-005D	US-UC-011C — CR uploads mid-term question paper	TC-UC-005D
FR-UC-005E	US-UC-011D — CR uploads final exam question paper	TC-UC-005E
FR-UC-006	US-UC-004 — Project showcase GitHub link	TC-UC-006
FR-UC-007	US-UC-016 — Bilingual search	TC-UC-016
FR-UC-008	US-UC-005 — Marketplace listing by verified user	TC-UC-007, TC-UC-008
FR-UC-009	US-UC-006 — Category filter in marketplace	TC-UC-009
FR-UC-010	US-UC-007 — Housing listing with map pin	TC-UC-010
FR-UC-011	US-UC-008 — CSE-Only roommate filter	TC-UC-011
FR-UC-012	US-UC-008 — Availability toggle real-time update	TC-UC-012
FR-UC-013	US-UC-009 — SOS panic button broadcast	TC-UC-013
FR-UC-014	US-UC-010 — Proctorial directory one-tap call	TC-UC-014
FR-UC-015	US-UC-013 — Admin content moderation queue	TC-UC-015
FR-UC-016	UC-03 — Upload type tagging validation	TC-UC-017
FR-UC-017	UC-03 — Duplicate upload detection	TC-UC-018

11.2 Initial Manual Test Plan

Test ID	Scenario	Preconditions	Steps	Expected Result
TC-UC-001	Non-university email rejected at registration	User at registration page	Enter non-@university.edu email; submit	Error: 'Must use university email'
TC-UC-002	FB-Auth mandatory — login fails without FB link	User registered without FB link	Attempt login; skip FB linking	Login blocked; FB linking prompt displayed
TC-UC-003	Resource navigation: Batch > Semester > Course > Type Tab	User logged in; resources uploaded	Navigate Archive > Select Batch > Semester > Course > click Slides tab	Only slide PDFs listed under selected course
TC-UC-004	Mid-term question bank year + tag filter	Mid-term papers uploaded with year tags	Select course tag 'CSE-301'; select year '2023'; select type 'Mid-Term'	Only CSE-301 mid-term papers from 2023 returned
TC-UC-004B	Final exam question bank year + tag filter	Final papers uploaded with year tags	Select course tag 'CSE-301'; select year '2023'; select type 'Final'	Only CSE-301 final papers from 2023 returned
TC-UC-005	CR uploads lab sheet; student downloads it	CR logged in with CR role	CR selects resource type 'Lab Sheet'; fills batch/semester/course; uploads PDF	File visible and downloadable after admin approval
TC-UC-005B	CR uploads lecture slide with lecture number tag	CR logged in with CR role	CR selects resource type 'Lecture Slide'; fills course + lecture no. = 5; uploads PDF	Slide appears under correct course Slides tab with lecture no. 5 displayed
TC-UC-005C	CR uploads reference book with author and edition	CR logged in with CR role	CR selects resource type 'Reference Book'; fills course, author='Knuth', edition='3rd'; uploads PDF	Book appears under course Books tab with author and edition metadata shown
TC-UC-005D	CR uploads mid-term question paper with year	CR logged in with CR role	CR selects type 'Mid-Term Paper'; fills course='CSE-301', year='2022'; uploads PDF	Paper appears in Question Bank under Mid-Term > CSE-301 > 2022
TC-UC-005E	CR uploads final exam question paper with year	CR logged in with CR role	CR selects type 'Final Paper'; fills course='CSE-401', year='2023'; uploads PDF	Paper appears in Question Bank under Final > CSE-401 > 2023
TC-UC-006	Project submission rejects invalid GitHub URL	CR on project showcase form	Submit project with URL: 'notgithub.com/repo'	Error: 'Must be a valid GitHub URL'

TC-UC-007	Marketplace contact info hidden from guest	Unauthenticated user views marketplace	Browse item; click 'Contact Seller'	Login prompt shown; seller info not revealed
TC-UC-008	Verified user can view seller contact info	Verified user views marketplace item	Click 'Contact Seller' on item listing	Seller FB Messenger link or contact info displayed
TC-UC-009	Marketplace hardware category filter	10+ items listed across categories	Select 'Hardware' filter	Only hardware items shown; no books or software
TC-UC-010	Housing listing map pin renders at correct coords	Housing post submitted with lat/lng	View housing listings in map mode	Pin appears at exact coordinates of submitted listing
TC-UC-011	CSE-Only filter hides non-CSE housing posts	Mixed CSE and non-CSE housing posts	Toggle 'CSE-Only' filter ON	Only is_cse_only=true listings visible
TC-UC-012	Availability toggle updates within 5 seconds	Housing listing with Available status	Lister toggles status to Occupied	Status change reflected for other users within 5 seconds
TC-UC-013	SOS alert reaches active users within 2 seconds	User A logged in with GPS; User B active nearby	User A holds Panic Button for 3 seconds	User B receives alert notification within 2 seconds
TC-UC-014	Proctorial directory loads and initiates call	User on any screen	Navigate to Safety > Directory; tap Security number	Directory loads in ≤2 seconds; phone dialer opens with number
TC-UC-015	Admin moderation: approve resource makes it visible	CR uploaded resource; Admin on moderation queue	Admin clicks 'Approve' on pending resource	Resource appears in student-facing archive under correct type tab immediately
TC-UC-016	Bilingual search returns same results in Bengali and English	Resources uploaded with both English and Bengali tags	Search for course name in Bengali script	Both queries return identical resource list
TC-UC-017	Upload rejected if required tag is missing	CR on upload form	CR selects 'Mid-Term Paper' type; leaves year field blank; submits	Error: 'Year is required for Mid-Term Paper uploads'
TC-UC-018	Duplicate upload warning shown	Identical file already uploaded	CR uploads file with same name + course + year as existing entry	Warning shown: 'This resource may already exist. Overwrite or Cancel?'

12. Lab 2 Submission Checklist

- Problem statement and scope are clear. (Section 2)
- Minimum 12 user stories with acceptance criteria. (Section 6.1 — 22 stories included)
- Functional requirements table is complete. (Section 5 — 19 requirements including all CR upload types)
- Data model and data dictionary are attached. (Section 7 — ER diagram + unified RESOURCE entity with type ENUM)
- Minimum 5 wireframes or flow screens included. (Section 8.1 — Use Case Diagram + 6 Navigation Flow Maps including CR Upload Flow)
- NFR section is completed. (Section 9 — 11 NFRs including storage quota)
- Risks and open questions are documented. (Section 10 — 8 risks, 7 open questions)
- Initial manual test plan (15+ cases) is included. (Section 11.2 — 23 test cases covering all upload types)